PHY101: Introduction to Physics-1

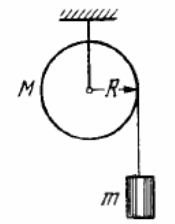
Monsoon, 2017

Tutorial-14

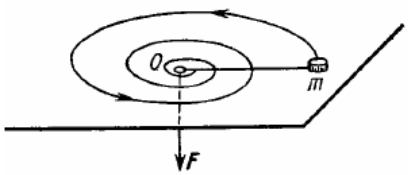
1. A wheel is attached to a fixed shaft, and the system is free to rotate without friction. To measure the moment of inertia of the wheel–shaft system, a tape of negligible mass wrapped around the shaft is pulled with a known constant force F. When a length L of tape has unwound, the system is rotating with angular speed ω_0. Find the moment of inertia I_0 of the system.

2. A sphere starting from rest rolls without slipping down an incline 1 in 8. Find the linear velocity of the sphere after it has traveled a distance of 1.6 m.*[Use the expression of moment of inertia of a uniform solid sphere.]*

3. A light thread with a body of mass m tied to its end is wound on a uniform solid cylinder of mass $\boldsymbol{M}$ and radius R (see the figure). At a moment $t = 0$ the system is set in motion. Assuming the friction in the axle of the cylinder to be negligible, find the time dependence of (a) the angular velocity of the cylinder; and (b) the kinetic energy of the whole system.



4. A small body of mass m tied to a non-stretchable thread moves over a smooth horizontal plane. The other end of the thread is being drawn into a hole *0* (see figure) with a constant velocity. Find the thread tension as a function of the distance r between the body and the hole if at $r = r_0$ the angular velocity of the thread is equal to ω_o.

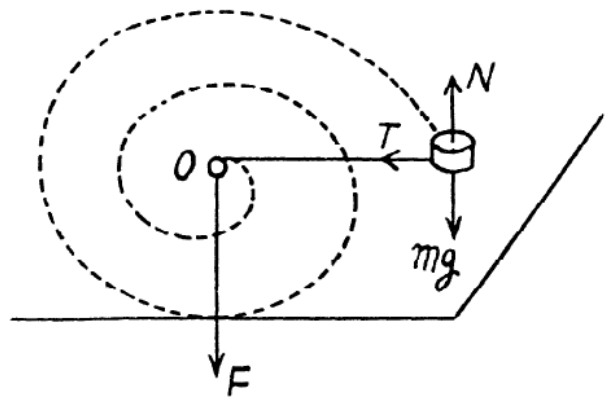


Forces, acting on the mass m are shown in the figure. As $\vec{N} = m\vec{g}$, the net torque of these two forces about any fixed point must be equal to zero. Tension T, acting on the mass m is a central force, which is always directed towards the centre O. Hence the moment of force T is also zero about the point O and therefore the angular momentum of the particle m is conserved about O.

Let, the angular velocity of the particle be ω, when the separation between hole and particle m is r, then from the conservation of momentum about the point O, :

$$m(\omega_0 r_0) r_0 = m(\omega r) r,$$

or $$\omega = \frac{\omega_0 r_0^2}{r^2}$$



Now, from the second law of motion for m,

$$T = F = m\omega^2 r$$

Hence the sought tension;

$$F = \frac{m\omega_0^2 r_0^4 r}{r^4} = \frac{m\omega_0^2 r_0^4}{r^3}$$